

NAYAN MAKANJEE

BSc Computer Science Student

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PROFILE

I am a 3rd-year Computer Science student at the University of Witwatersrand with hands-on full-stack and machine learning experience across shipped or in development projects as well as technical support, and problem-solving across multiple domains. The projects include a live student marketplace and a coaching platform with offline-sync support. Seeking opportunities to apply programming skills, analytical thinking, and a collaborative mindset in an entry-level IT or software role. Known for diligence, adaptability, and a drive to continuously grow.

TECHNICAL SKILLS

Languages	Python, Java, C++, C#, Javascript, TypeScript, SQL
Frontend / Web	HTML, CSS, React
Backend & Data	Node.Js, Next.JS, Supabase, PostgreSQL, Drizzle ORM, REST APIs
Data Science & ML	NumPy, Pandas, Scikit-learn, LightGBM
Tools / Development	Git, GitHub, VS Code, Android Studio, AutoCAD, Netbeans
Concepts	OOP, Data Structures, Algorithms, Zod, Agile/Scrum, CI/CD Pipelines
Office / OS	Microsoft Office Suite, Windows, Linux (basic)

PROJECTS

CampusXChange

2026

University of Witwatersrand: Team of 4 <https://mango-bay-0edf29110.7.azurestaticapps.net/>
<https://github.com/nayan-m15/Campus-Marketplace>

- This website serves as a secondhand marketplace for students to buy, sell or trade items safely within university context, with separate admin and staff dashboards to manage facilities, bookings, and marketplace operations.
- Stack: React, Vite, CSS, Supabase, jspdf, Azure static web apps, PayFast, Brevo and SerpAPI.
- Used Git for version control.

Gaffer – In Development

2026

University of Witwatersrand: Team of 6 <https://gaffer-irid.vercel.app/>
<https://github.com/nayan-m15/Gaffer>

- This is a full-stack coaching platform for amateur football teams to manage rosters, schedule fixtures, and record live match events, with player statistics automatically derived from event logs.
- The offline integration is done using PowerSync with background synchronisation and duplicate-entry detection when data is merged across devices.
- The backend is built with NestJS, Drizzle ORM, and Neon PostgreSQL, and the frontend with React 19, TypeScript, TanStack Query, and shadcn/ui.
- Implementing role-based access control for Coach, Assistant, and Athlete users, with Zod-validated APIs documented using Swagger/OpenAPI.

Human Activity Recognition

2026

University of Witwatersrand: Team of 4
Machine Learning Group Project

- Developed an end-to-end machine learning pipeline to identify six human activities from multivariate time-series data collected through a wearable sensor system.
- Engineered a robust preprocessing workflow using multi-stage missing-data imputation, combining interpolation, sample-level statistical filling, and zero replacement to eliminate sensor gaps across the dataset.
- Extracted over 1,470 features per sample, incorporating statistical measures, frequency-domain FFT features, autocorrelation patterns, temporal phase segmentation, and cross-sensor correlations.
- Evaluated multiple classification approaches using stratified 5-fold cross-validation, selecting a tuned LightGBM model that achieved a Macro-F1 score of 0.797 ± 0.021 .
- Technologies: Python, NumPy, pandas, scikit-learn, LightGBM; developed as a Kaggle competition submission.

Crafted Interiors

2026

Carpentry Business Website <https://www.craftedinteriors.co.za/>

- Contributed to the development and refinement of a professional website for a bespoke carpentry and interiors company, working within an existing React codebase.
- Redesigned the About page and reworked the hero section video to improve the overall user experience.
- Tech Stack: React, TypeScript, Vite and Tailwind CSS

GrubFlow

2025

University of Witwatersrand: Team of 4
https://github.com/nayan-m15/Project_GrubFlow

- This is a restaurant order tracking android application which lets staff track orders in real time.
- Built with Android Studio with a PHP backend.
- Features: User Login, Live Order Status Updates and an Admin Dashboard
- This was a purely academic application and was never used by actual staff members of a restaurant.

EXPERIENCE & VOLUNTEERING

Head of Technical Support

2022 – 2023

Curro Midrand Sagewood Campus

- Managed all lighting, sound and IT equipment setup for school events, assemblies, and functions, diagnosing and resolving hardware/software faults under time pressure.
- Trained and supervised junior students in independently operating presentation and sound systems.

Volunteer

2021 – 2023

Food for Life & Sandton SPCA

- Contributed regular volunteer hours to community food distribution and animal welfare.

EDUCATION

BSc Computer Science

2024 – Present

University of Witwatersrand

- Relevant coursework: Software Design, Operating Systems & Systems Programming, Computer Graphics & Visualization, Data Structures & Algorithms, Object-Oriented Programming, Databases, Networks, Machine Learning, Parallel Computing.

National Senior Certificate (Matric)

2023

Curro Midrand Sagewood Campus

- Honours Distinction; Full Colours for Academics: Grades 8 – 11; Certificate in Technical Support
- Prefect: Grade 7 (2018) & Grade 12 (2023)

ACHIEVEMENTS & ACTIVITIES

- Captained school's cricket side and competed at regional trials (2014, 2015, 2018)
- English Public Speaking: 2019 – 2022; Vega Vibe Design Challenge Participant (2022)
- FSKA Karate World Championship participant (2015); SAICI JHB Bridge Building Competition participant

INTERESTS

Guitar · Software development · Hiking & outdoor adventure · Gaming · Fitness & Sport